JONATHAN PARLETT

(484) · 502 · 4520 ◇ jp86440@uga.edu
Website ◇ https://jmparlett.github.io

EDUCATION

University of Georgia *2025-Present*
Ph.D. Mathematics

Drexel University *2020-2025*
B.S. Mathematics
B.S. Computer Science
GPA: 3.8

RESEARCH INTERESTS

Arithmetic Geometry, Algebra, and Combinatorics.

PUBLICATIONS

2025 Top to random and reverse: analysis of a new descent algebra shuffle, joint with Darij Grinberg. Preprint. [arXiv]

2025 Fixed Point Homing Shuffles. *Discrete Mathematics & Theoretical Computer Science*, vol. 27:1. [DOI]

2023 Subnet Communicability: Diffusive Communication Across the Brain Through a Backbone Subnetwork, joint with Abhishek Jeyapratap, Ali Shokoufandeh, Birkan Tunc, and Yusuf Osmanlioglu. *CDMRI 2023*. [Springer]

PRESENTATIONS/TALKS

2025 Long Paths in Polynomial Divisor Graphs, joint with Nicole Froitzheim. Poster, Joint Mathematics Meetings. [poster]

2024 Long Paths in Polynomial Divisor Graphs, joint with Kayla Traxler. YMC 2024, Ohio State University. [slides]

2023 Subnet Communicability, joint with Abhishek Jeyapratap. CDMRI 2023. [slides]

2023 Subnet Communicability. Drexel Week of Undergraduate Excellence. [video]

TEACHING EXPERIENCE

University of Georgia

Instructor of Record
Pre-calculus Math 1113: Fall 2026

Grader
Applied Linear Algebra: Fall 2025
Introduction to Higher Mathematics: Spring 2026
Graduate Algebra I: Fall 2026

Drexel University I worked as a tutor and grader in the computer science and mathematics departments during undergraduate. I supported various courses such as Data Structures and Algorithms,

Programming Languages, Algorithmic Number Theory & Cryptography, Abstract and Linear Algebra. Responsibilities included holding office hours, grading, and occasionally creating supplemental course material.

PROFESSIONAL EXPERIENCE

SRI International March - September 2022
Intern, Data Analysis and Software Engineering *Princeton, NJ*

Worked under Principal Research Engineer Norman Hurst, to develop a technology to allow for recovery of time from video streams, robust against framerate conversions.

Conducted statistical analysis of system performance to facilitate improvement and validation of functionality, as well as develop methods to further improve accuracy of this system. Contributions led to inclusion as co-inventor on patent.

Worked primarily in the C programming language, with performance analysis through python and pandas.

United States Marine Corps October 2015 - February 2020
Data Systems Administrator *Camp Pendleton, CA*

Configured and maintained information systems networks to facilitate communication between military units in remote locales.

Provided services such as voice, video, email, file sharing, and internet, through a combination of Cisco, and Microsoft software and hardware solutions.

Trained peers and subordinates in occupational skills.

AWARDS

Robert J. Bickel Endowed Scholarship, Spring 2025

Outstanding Undergraduate Teaching Assistant Award, Spring 2024

Yilin Yang Outstanding Undergraduate Math Research Award, Spring 2024

Undergraduate Research Mini-Grant, Winter 2023

Dean's List, Drexel University

Navy and Marine Corps Achievement Medal, United States Marine Corps 2018, 2020